AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM352
Web, Mobile and Cloud Technologies

Midterm Examination (MTA)
Fall Semester 2025/2026

Date: 19/Nov./2025

Number of Exam Pages: 3 (including this cover sheet)	Time Allowed: 2 Hrs

Instructions:

- Total Marks: 45 {will be adjusted to 30 marks}
- This exam consists of 3 parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are not allowed.

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PART I (5 Marks)

Answer all the questions

Choose the right answer

1) What are the three key objectives at the heart of computer security according to the CIA triad?
A) Confidentiality, Integrity, and Authentication
B) Confidentiality, Integrity, and Availability
C) Confidentiality, Identification, and Authorization
D) Cryptography, Integrity, and Availability

2) In symmetric key encryption methods, what is the main challenge or problem?
A) They require too much computation power
B) They are not secure against brute force attacks
C) Both parties need to have the same key, and the key exchange must happen in advance
D) They can only encrypt small amounts of data

3) What is a key advantage of JSON over XML?
A) JSON supports more data types than XML
B) JSON is more compact and maps easily to JavaScript objects without requiring a parser
C) JSON is better for large documents
D) JSON has stricter validation rules

4) _____ distinguish between different applications running on any one computer in a network.
A) TCP Protocol
B) TCP IP Protocol
C) TCP Port number
D) TCP FTP Protocol

5) web page element _____ can contain other elements
A) select
B) input
C) button
D) label

PART II (20 Marks)

Answer all the questions

1. List and briefly explain THREE new features introduced in HTML5 that were not available in previous HTML versions. [6 Marks]
2. Discuss Open standards and the benefits of engaging in open standards.[5 Marks]
3. Define the **Document Object Model (DOM)** and explain its purpose in web development. [4 Marks]
4. Compare between SOAP and RESTful web services in terms of service transporting protocols and messaging protocols for both request and response messages.[5 Marks]

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PART III (20 Marks)

Answer all the questions

1. Provide JSON Object representation of the given data [6 Marks]

Time	Flight	From	Airline	Aircraft
9 AM	MS966	China	Egypt air	A321
10 AM	MS950	Spain	Spain air	A3201
11 AM	MS1023	London	Lufthansa	A777

2. Complete the missing annotations for the given RESTful web service [5 marks]

The following is a RESTful web service called "CaseConversion" that has a services called "toLower" to convert a lower case string to upper case.

```
..... ("case")
public class CaseConversion
{
   .....
   .....("/lower/{word}")
   .....(MediaType.TEXT_HTML)
   public String toLower(..... ("word") String word)
   {
      return word.toLowerCase();
   }
```

3. Write an HTML page with a form containing two mandatory textfields for student name and his/her Date of Birth. On submission, the values are sent to the URI "/collecting" using post method. [6 marks]

4. Complete the HTTP request that would be generated when a page www.historyNovels.com/p1.html is typed in the address part [3 marks]

```
..... ..... HTTP/1.1
   Host: .....
```

End of questions

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